

# Writing Operators as Sum of Commutators

Yizhao Chen

September 8, 2026

This note is a summary of some parts of the paper *Commutators in Ideals of Trace Class Operators* by Anderson and Vaserstein. We will not present all the details (since they can be found in the original paper), but will only give the key ideas of their proof and some applications of the results. To avoid ambiguity, from now on the set of natural numbers  $\mathbb{N}$  includes 0.

## Contents

|          |                                        |           |
|----------|----------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                    | <b>1</b>  |
| <b>2</b> | <b>A Simple Example</b>                | <b>3</b>  |
| <b>3</b> | <b>Improving the Lemma</b>             | <b>4</b>  |
| <b>4</b> | <b><math>(N, p)</math>-Summability</b> | <b>6</b>  |
| <b>5</b> | <b>Proof of the Main Theorem</b>       | <b>8</b>  |
| <b>6</b> | <b>Applications</b>                    | <b>10</b> |

## 1 Introduction

Understanding the motivation of Anderson and Vaserstein requires additional knowledge<sup>1</sup>. For this reason, we use a simpler problem as motivation: for an  $n \times n$  matrix

---

<sup>1</sup>To be precise, one needs to know some algebraic K-theory.

$A$  over  $\mathbb{C}$ , when can it be written as a sum of commutators, i.e.

$$A = \sum_{k=1}^N [B_k, C_k], \quad [B_k, C_k] = B_k C_k - C_k B_k?$$

This question has a very simple answer:

**Theorem 1.1.** *If a matrix  $A \in M_n(\mathbb{C})$  has trace  $\text{Tr } A = 0$ , then it can be written as a finite sum of commutators.*

The proof of this theorem is quite elementary. Roughly speaking, it suffices to show that matrices of the forms

$$\begin{pmatrix} d & 0 \\ 0 & -d \end{pmatrix} \quad \begin{pmatrix} 0 & a \\ 0 & 0 \end{pmatrix}$$

can be expressed as sums of commutators. This can be found in many sources, such as Anderson-Vaserstein's paper.

A natural generalization is: for a bounded operator  $A \in \mathcal{L} = \mathcal{L}(\mathcal{H})$  on an infinite-dimensional separable Hilbert space  $\mathcal{H}$ , when can it be expressed as a sum of commutators? Similar to the previous problem, one might guess that there is a theorem stating that if an operator has trace zero, then it can be written as a sum of commutators.

Such a naive idea faces many issues. The first is that we cannot define the trace for all operators. This leads to the notion of trace-class operators. In general, we need to discuss the so-called Schatten  $p$ -ideals

$$\mathcal{L}^p = \{A \in \mathcal{L} \mid \text{Tr} \left( (A^* A)^{\frac{p}{2}} \right) < +\infty\}, \quad 0 < p \leq +\infty.$$

We omit the discussion of Schatten  $p$ -ideals here; the properties we need can be found in Simons' *Trace Ideals and Their Applications*.

Another issue is that if we consider the most general commutators, the desired theorem becomes trivial. Using the result of Brown and Pearcy in *Structure of commutators of operators*, any bounded linear operator can be written as a sum of at most two commutators. Thus we restrict ourselves to commutators of trace-class operators with other operators, i.e.

$$[\mathcal{L}^1, \mathcal{L}] = \text{Span}\{[T, A] \mid T \in \mathcal{L}^1, A \in \mathcal{L}\}.$$

We note that in finite dimensions we have  $\mathcal{L}^1 = \mathcal{L}$ , so the above space reduces to the space spanned by commutators.

Our main theorem is therefore:

**Theorem 1.2.** *For  $0 < p < 1$ , if an operator  $A \in \mathcal{L}^p$  satisfies  $\text{Tr } A = 0$ , then  $A \in [\mathcal{L}^1, \mathcal{L}]$ .*

**Remark 1.1.** *If we weaken the condition to  $A \in \mathcal{L}^1$  and  $\text{Tr } A = 0$ , the theorem is no longer true. A concrete counterexample can be found in the paper of Anderson and Vaserstein.*

## 2 A Simple Example

The key idea of the proof of Anderson and Vaserstein comes from the analysis of the following simple example:

**Example 2.1.** *Consider the Hilbert space  $\mathcal{H} = l^2(\mathbb{Z})$ . Any bounded function on  $\mathbb{Z}$  acts on it by pointwise multiplication, so any bounded function  $\sigma(n)$  can be viewed as a bounded operator on  $\mathcal{H}$ . Consider the shift operator*

$$T: f(\cdot) \mapsto f(\cdot + 1).$$

*One can verify that*

$$([\sigma T^{-1}, T]f)(n) = ((\sigma - T\sigma T^{-1})f)(n) = (\sigma(n) - \sigma(n+1))f(n).$$

*Therefore, for any bounded function  $\lambda$ , if we set  $\sigma(n) - \sigma(n+1) = \lambda(n)$  and  $\sigma(0) = 0$ , we can find some (possibly unbounded) operator  $A = \sigma T^{-1}$  such that*

$$\lambda = [A, T].$$

*Moreover, if we can show that  $\sigma \in \mathcal{L}^1$ , then  $\lambda \in [\mathcal{L}^1, \mathcal{L}]$ .*

In this example, it is easy to write down the equivalent condition for  $\sigma$  to be a trace-class operator:  $\sum_{n \in \mathbb{Z}} |\sigma(n)| < +\infty$ . Since it is also not hard to express  $\sigma$  in terms of  $\lambda$ , after imposing additional conditions on  $\lambda(n)$ , up to unitary conjugacy we obtain the following lemma:

**Lemma 2.1.** *Let a normal trace-class operator  $A$  have eigenvalues  $\{\lambda_n\}_{n \in \mathbb{N}}$  (with multiplicity, i.e. repeated eigenvalues are listed multiple times). If*

$$\sum_{n=1}^{+\infty} \left| \sum_{k=n}^{\infty} \lambda_k \right| < +\infty$$

*and  $\text{Tr } A = 0$ , then  $A \oplus 0 \in \mathcal{L}(\mathcal{H} \oplus \mathcal{H})$  lies in  $[\mathcal{L}^1(\mathcal{H}), \mathcal{L}(\mathcal{H})]$ .*

*Proof.* Since  $A$  is compact and normal, the spectral theorem for normal operators gives an orthonormal basis  $\{e_n\}_{n \in \mathbb{N}}$  of  $\mathcal{H}$  such that  $Ae_n = \lambda_n e_n$ . We can extend this to an orthonormal basis  $\{e_n\}_{n \in \mathbb{Z}}$  of  $\mathcal{H} \oplus \mathcal{H}$ . Using this basis, we identify  $\mathcal{H} \oplus \mathcal{H}$  with  $l^2(\mathbb{Z})$ . Correspondingly,  $A$  is unitarily equivalent to the operator

$$(\hat{A}f)(n) = \begin{cases} \lambda_n f(n) & n \geq 0, \\ 0 & n < 0. \end{cases}$$

Now consider the function  $\sigma$  from the previous example. We obtain

$$\sigma(n) = \begin{cases} \sum_{k=n}^{+\infty} \lambda_k & n \geq 0, \\ 0 & n < 0. \end{cases}$$

The value of  $\sigma$  on non-positive integers is 0 because  $\text{Tr } A = 0$ , i.e.  $\sum_{n=1}^{\infty} \lambda_n = 0$ . Combining the definition of  $\sigma$  with the condition in the lemma, we know that  $\sigma$  defines a trace-class operator, and hence  $\hat{A} = [\sigma T^{-1}, T]$  lies in  $[\mathcal{L}^1(l^2(\mathbb{Z})), \mathcal{L}(l^2(\mathbb{Z}))]$ . Using the isometric isomorphism between  $\mathcal{H} \oplus \mathcal{H}$  and  $l^2(\mathbb{Z})$ , we obtain the desired result.  $\square$

### 3 Improving the Lemma

This lemma is still a bit far from our main theorem. For example, consider the operator on  $l^2(\mathbb{Z})$  defined by the function

$$\lambda(n) = \begin{cases} \frac{1}{1+|n|^{\frac{3}{2}}} & n \neq 0, \\ -2 \sum_{k=1}^{+\infty} \frac{1}{1+|k|^{\frac{3}{2}}} & n = 0 \end{cases}$$

It lies in  $\mathcal{L}^{3/4}$  and has trace zero, but it is easy to check that it does not satisfy the condition of the lemma. To prove the main theorem, we need to improve this method slightly. First, mimicking the finite-dimensional result, we can prove:

**Lemma 3.1.** *If  $N$  trace-class operators  $A_1, A_2, \dots, A_N$  satisfy  $\sum_{n=1}^N A_n = 0$ , then  $A_1 \oplus A_2 \oplus \dots \oplus A_N$  lies in  $[\mathcal{L}^1, \mathcal{L}]$ .*

This allows us to relax the requirements in the previous example: we want to construct an equality of the form

$$[\sigma S, S^{-1}] = A + A_1 \oplus \dots \oplus A_N,$$

where  $S$  is some permutation of the basis and  $A_1, \dots, A_N$  are trace-class operators with sum zero.

Specifically, let  $\sigma(n)$  be a bounded function on  $\mathbb{Z}$  such that

$$\sigma(n) = 0, \quad \forall n \leq 0,$$

acting on  $l^2(\mathbb{Z})$  by pointwise multiplication. Choose the injection

$$\begin{aligned} S: \mathbb{N} &\longrightarrow \mathbb{N} \\ n &\longmapsto Nn + 1 \end{aligned}$$

and extend it to a bijection from  $\mathbb{Z}$  to itself. This bijection gives a unitary operator on  $l^2(\mathbb{Z})$ :

$$(Sf)(n) = f(S(n)), \quad \forall f \in l^2(\mathbb{Z}), n \in \mathbb{Z}.$$

We can compute:

$$((\sigma - S^{-1}\sigma S)f)(n) = \begin{cases} (\sigma(n) - \sigma(\frac{n-1}{N}))f(n) & n = Nm + 1; \\ \sigma(n)f(n) & n \neq Nm + 1. \end{cases}$$

Now, for any normal operator  $A$  with eigenvalues  $\{\lambda_n\}_{n \in \mathbb{N}}$ , if we can find  $\sigma(n)$  such that:

$$\begin{aligned} \sigma(1) - \sigma(0) + \sigma(2) + \sigma(3) + \dots + \sigma(N) &= \lambda_0 \\ \sigma(N+1) - \sigma(1) + \sigma(N+2) + \sigma(N+3) + \dots + \sigma(N+N) &= \lambda_1 \\ \sigma(2N+1) - \sigma(2) + \sigma(2N+2) + \sigma(2N+3) + \dots + \sigma(2N+N) &= \lambda_2 \\ &\vdots \\ \sigma(mN+1) - \sigma(m) + \sigma(mN+2) + \sigma(mN+3) + \dots + \sigma(mN+N) &= \lambda_m \\ &\vdots \end{aligned} \tag{1}$$

then, by grouping the basis indices modulo  $N$ , we define subspaces

$$\mathcal{H}_n = \text{closure of Span}\{\delta_{mN+n} \mid m \in \mathbb{Z}\}, \quad n = 1, 2, \dots, N,$$

where  $\delta_m$  is the function that is 1 at  $m$  and 0 elsewhere. We then define operators

$$\hat{A}\delta_n = \begin{cases} \lambda_m \delta_n & n = mN, \\ 0 & n \neq mN \end{cases}$$

and<sup>2</sup>

$$\begin{aligned} A_1 &= (\sigma - S^{-1}\sigma S)|_{\mathcal{H}_1} - \hat{A}, \\ A_n &= (\sigma - S^{-1}\sigma S)|_{\mathcal{H}_n}, \quad 1 < n \leq N. \end{aligned}$$

Thus we obtain the desired equality:

$$\begin{aligned} [\sigma S, S^{-1}] &= \hat{A} + A_1 \oplus A_2 \oplus \dots \oplus A_N, \\ A_1 + A_2 + \dots + A_N &= 0, \end{aligned}$$

where the second equality is understood after suitably identifying each  $\mathcal{H}_n$  with  $\mathcal{H}$ .

Note that  $\hat{A}$  is unitarily equivalent to  $A \oplus 0$ . Hence it remains to find an appropriate  $\sigma(n)$  and give sufficient conditions for  $\sigma$  and  $A_1, \dots, A_N$  to be trace-class operators.

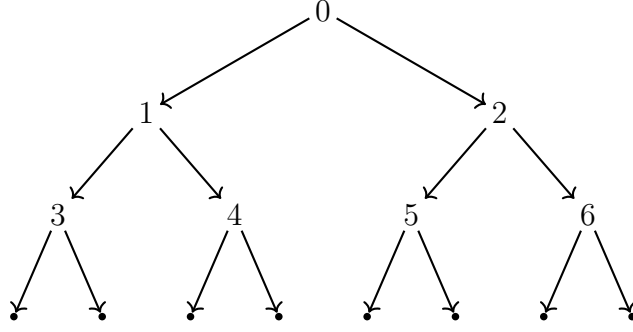
## 4 $(N, p)$ -Summability

For a fixed sequence of complex numbers  $\{\lambda_n\}_{n \in \mathbb{N}}$ , let us first find a suitable function  $\sigma$  satisfying condition (1). For convenience, we assume  $\sum_n \lambda_n = 0$  (which corresponds to the trace-zero condition). We also assume  $N = 2$ ; for general  $N$ , the construction of  $\sigma(n)$  is similar.

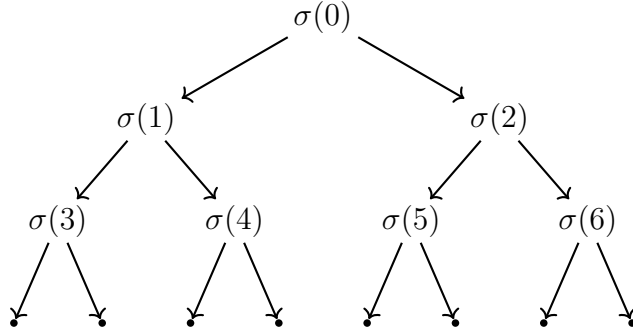
Let us view the above condition from another perspective: place  $\mathbb{N}$  on a rooted tree  $\mathcal{T}$ , where each node  $m$  points to exactly  $N$  nodes  $mN+1, mN+2, \dots, mN+N$ . The tree for  $N = 2$  is shown below:

---

<sup>2</sup>We can imagine arranging the eigenvalues of  $\sigma - S^{-1}\sigma S$  into a table with  $N$  columns. The previous conditions then say that the sum of each row of the table is  $\lambda_n$ , and extracting each column gives  $\hat{A}$  and  $A_1, \dots, A_N$ .



We say that a node  $m$  is a descendant of another node  $n$  if they are equal or if there is a directed path from  $n$  to  $m$ , and denote this relation by  $n \succeq m$ . If we replace each node  $m$  in the tree by  $\sigma(m)$ , we get:



Considering one node individually, condition (1) gives:

$$\begin{array}{c} \sigma(2m+1) + \sigma(2m+2) - \lambda_m \\ \swarrow \quad \searrow \\ \sigma(2m+1) \quad \sigma(2m+2) \end{array}$$

In other words, the value of  $\sigma$  at each node  $m$  is larger than the sum of the values at its children by  $-\lambda_m$ . Such a  $\sigma$  is not hard to find: for each  $n \in \mathbb{N}$ , take all its descendants  $m$ , and set  $\sigma(n)$  to be the sum of the corresponding  $-\lambda_m$ 's, i.e.

$$\sigma(n) = - \sum_{m \succeq n} \lambda_m.$$

One can verify that such  $\sigma(n)$  satisfies condition (1).

It is also easy to see that the condition for this  $\sigma$  to be a trace-class operator is exactly

$$\sum_{n=1}^{+\infty} \left| \sum_{m \succeq n} \lambda_m \right| < +\infty. \quad (2)$$

More generally, for a tree  $\mathcal{T}_N$  determined by some  $N > 1$ , the condition for  $\sigma$  defined from  $\lambda_m$  to lie in  $\mathcal{L}^p$  is

$$\sum_{n=1}^{+\infty} \left| \sum_{m \succeq_N n} \lambda_m \right|^p < +\infty.$$

This condition is called  $(N, p)$ -summability by Anderson and Vaserstein. The main reason for introducing the parameter  $p$  is that Anderson and Vaserstein wished to discuss more general commutators  $[\mathcal{L}^p, \mathcal{L}]$ , and one way to determine whether an operator lies in it is by using  $(N, p)$ -summability. In fact, the conclusions and proofs in this note can be generalized to general  $0 < p \leq p$ . Details can be found in their paper.

It is not hard to verify that when a sequence  $\{\lambda_n\}_{n \in \mathbb{N}}$  satisfies  $(N, 1)$ -summability, the operators  $A_1, \dots, A_N$  defined in the previous section are all trace-class. Combining all the discussion so far, we obtain:

**Lemma 4.1.** *If there exists some  $N > 1$  such that the eigenvalues  $\{\lambda_n\}_{n \in \mathbb{N}}$  of a normal trace-class operator  $A$  satisfy the  $(N, 1)$ -summability condition (2), then  $A \oplus 0$  lies in  $[\mathcal{L}^1, \mathcal{L}]$ .*

## 5 Proof of the Main Theorem

So far, we have only been able to prove that operators of the form  $A \oplus 0$  lie in  $[\mathcal{L}^1, \mathcal{L}]$ . In this section we extend these results to arbitrary normal operators, i.e. we prove:

**Theorem 5.1.** *If there exists some  $N > 1$  such that the eigenvalues  $\{\lambda_n\}_{n \in \mathbb{N}}$  of a normal trace-class operator  $A$  satisfy the  $(N, 1)$ -summability condition (2), then  $A$  lies in  $[\mathcal{L}^1, \mathcal{L}]$ .*

Using this theorem, the main theorem follows readily.

To prove this theorem, we only need the following lemma and the earlier Lemma 3.1:

**Lemma 5.1.** *Let a normal trace-class operator  $A$  have eigenvalues satisfying  $(N, 1)$ -summability. Then there exists a finite-rank operator  $F$  and  $N$  normal trace-class operators  $B_1 \oplus 0, B_2 \oplus 0, \dots, B_N \oplus 0$  such that:*



1.  $A = F + (B_1 \oplus 0) + (B_2 \oplus 0) + \cdots + (B_N \oplus 0)$ ;
2.  $\text{Tr } F = \text{Tr } A$ ;
3.  $F, B_1, B_2, \dots, B_N$  all satisfy  $(N, 1)$ -summability and  $\text{Tr } B_n = 0$ ;
4. the zero summand in each  $B_n \oplus 0$  acts on an infinite-dimensional subspace.

We omit the proof of this lemma, only mentioning one thing: to preserve  $(N, 1)$ -summability, the construction uses the tree  $\mathcal{T}_N$ .

At this point we have an almost complete theorem; the only defect is that verifying  $(N, 1)$ -summability is somewhat cumbersome. Therefore, at the end of this section we give a sufficient condition for  $(N, 1)$ -summability.

**Lemma 5.2.** *If a sequence of complex numbers  $\{\lambda_n\}_{n \in \mathbb{N}}$  satisfies*

$$\sum_{n=1}^{+\infty} (\log n) |\lambda_n| < \infty,$$

*then for every  $N > 1$ , it satisfies  $(N, 1)$ -summability.*

*Sketch of proof.* It suffices to establish the inequality

$$\sum_{n=1}^{+\infty} \left| \sum_{m \succeq n} \lambda_m \right| \leq C \sum_{n=1}^{+\infty} (\log n) |\lambda_n|.$$

Simply enlarge the sum inside the absolute value to  $\sum_{m \succeq n} |\lambda_n|$ , and then count how many times each  $|\lambda_n|$  appears in the sum. The number of occurrences of each  $|\lambda_n|$  is roughly the number of ancestors of  $n$  in the tree  $\mathcal{T}_N$ , which is about  $\log_N n$ . This yields the right-hand side.  $\square$

Finally we give the proof of the main theorem. Suppose  $A \in \mathcal{L}^p$  with  $p < 1$  satisfies  $\text{Tr } A = 0$ . Since  $\mathcal{L}^p$  is closed under taking adjoints, we have  $A^* \in \mathcal{L}^p$ , and  $\text{Tr } A^* = \overline{\text{Tr } A} = 0$ . Therefore,

$$\Re A = \frac{1}{2}(A + A^*), \quad \Im A = \frac{1}{2i}(A - A^*)$$

both lie in  $\mathcal{L}^p$ , and both have trace zero. It suffices to show that  $\Re A$  and  $\Im A$  lie in  $[\mathcal{L}^1, \mathcal{L}]$ .

Let the eigenvalues of  $\Re A$  be  $\{\lambda_n\}_{n \in \mathbb{N}}$ , with  $\{|\lambda_n|\}$  decreasing in  $n$ . Since  $\Re A \in \mathcal{L}^p$ , we have

$$\sum_{n \in \mathbb{N}} |\lambda_n|^p < +\infty.$$

Together with the fact that  $|\lambda_n|$  is decreasing, this implies  $n|\lambda_n|^p \rightarrow 0$ . Hence

$$\lim_{n \rightarrow +\infty} \frac{(\log n)|\lambda_n|}{|\lambda_n|^p} = \lim_{n \rightarrow +\infty} \frac{\log n}{n^{\frac{1-p}{p}}} \cdot (n|\lambda_n|^p)^{\frac{1-p}{p}} = 0.$$

By the comparison test,  $\sum_{n=1}^{+\infty} (\log n)|\lambda_n|$  converges. Then by Lemma 5.2 and Theorem 5.1, we have  $\Re A \in [\mathcal{L}^1, \mathcal{L}]$ . The same argument applies to  $\Im A$ , completing the proof of the main theorem.

## 6 Applications

The results of Anderson and Vaserstein give many interesting representations of commutators. Let  $\mathcal{F}$  denote the set of all finite-rank operators. For two invertible operators  $U, V$ , denote their multiplicative commutator  $UVU^{-1}V^{-1}$  by  $\{U, V\}$ . For two subsets  $\mathcal{X}, \mathcal{Y} \subseteq \mathcal{L}$ , write

$$\{\mathcal{X}, \mathcal{Y}\} = \left\{ \prod_{k=1}^r \{X_k, Y_k\} \mid X_k \in \mathcal{X}, Y_k \in \mathcal{Y}, r \in \mathbb{N} \right\},$$

$$\mathcal{X} + \mathcal{Y} = \{X + Y \mid X \in \mathcal{X}, Y \in \mathcal{Y}\}.$$

Then we have:

**Theorem 6.1.**

$$\{1 + \mathcal{L}^1, \mathcal{L}\} + \mathcal{F} = 1 + [\mathcal{L}^1, \mathcal{L}] + \mathcal{F}.$$

*Here equality means equality of sets.*

Proving the inclusion from the left-hand side to the right-hand side uses properties of multiplicative commutators, so we omit the whole proof.

Consider the map

$$\begin{aligned} \varphi: (1 + \mathcal{L}^1)^\times &\longrightarrow \mathcal{L}^1/[\mathcal{L}^1, \mathcal{L}] \\ 1 + K &\longmapsto [K + F_K], \end{aligned}$$

where  $(1 + \mathcal{L}^1)^\times$  denotes the group of invertible elements in  $1 + \mathcal{L}^1$ , and  $F_K$  is any finite-rank operator satisfying  $\text{Tr}(K + F_K) = 0$ . Using the previous theorem, we obtain:

**Theorem 6.2.** *The definition of  $\varphi$  is independent of the choice of  $F_K$ , and it gives a homomorphism from the multiplicative group  $(1 + \mathcal{L}^1)^\times$  to the additive group  $\mathcal{L}^1/[\mathcal{L}^1, \mathcal{L}]$ . Moreover,  $\varphi$  induces an isomorphism*

$$\begin{aligned} Z_p &:= \{1 + K \in (1 + \mathcal{L}^1)^\times \mid \det(1 + K) = 1\} / \{1 + \mathcal{L}^1, \mathcal{L}\} \\ &\cong \{K \in \mathcal{L} \mid \text{Tr } K = 0\} / [\mathcal{L}^1, \mathcal{L}]. \end{aligned}$$

Readers familiar with algebraic K-theory will recognize that the group  $Z_p$  is actually a subgroup of the algebraic  $K_1$  group  $K_1^{alg}(\mathcal{L}, \mathcal{L}^1)$ <sup>3</sup>. Indeed, we have:

$$K_1^{alg}(\mathcal{L}, \mathcal{L}^1) \cong \mathbb{C} \oplus Z_p.$$

Therefore, the above conclusion gives another representation of  $K_1^{alg}(\mathcal{L}, \mathcal{L}^1)$ , namely

$$K_1^{alg}(\mathcal{L}, \mathcal{L}^1) \cong \mathbb{C} \oplus \ker \text{Tr} / [\mathcal{L}^1, \mathcal{L}].$$

In fact, the original motivation of Anderson and Vaserstein for considering commutators was precisely to determine the relative  $K_1$  group defined by the ideal  $\mathcal{L}^p$ .

---

<sup>3</sup>In fact, one also needs the fact that  $\mathcal{L}$  is of stable rank 1.